

Course code: CSA1013

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C01 ASSESSMENT TOOL 2

AGILE SPRINT PLANNING & USER STORY WORKSHOP

AI-Driven Smart Urban Noise Pollution Monitoring and Mitigation System

1. Project Vision & Stakeholder Identification

Project Vision:

The vision of the **AI-Driven Smart Urban Noise Pollution Monitoring and Mitigation System** is to develop an intelligent software solution that continuously monitors environmental noise levels using IoT sensors, Machine Learning, GIS, and cloud technologies. The system aims to detect noise pollution hotspots, generate real-time alerts, predict future noise trends, and assist government authorities in making informed decisions for reducing urban noise pollution. This solution supports smart city initiatives by improving environmental sustainability, public health, and regulatory compliance.

Project Objectives

The main objectives of the project are:

1. Monitor urban noise levels continuously.
2. Identify noise hotspots using GIS mapping.
3. Predict future noise trends using Machine Learning.
4. Generate real-time alerts when permissible noise levels are exceeded.
5. Analyze major sources of urban noise pollution.
6. Generate environmental compliance reports.
7. Support urban planning and policy-making.
8. Improve public health and quality of urban life.

Stakeholder Identification

Stakeholder	Role
Municipal Corporation	Monitors city-wide noise levels and takes corrective actions.
Pollution Control Board	Ensures industries and public areas comply with environmental noise standards
Citizens	Receive alerts and benefit from reduced noise pollution.
Industries & Commercial Areas	Monitor their operational noise and comply with regulations.
Traffic Police	Control traffic congestion and reduce traffic-related noise.
Hospitals & Educational Institutions	Ensure quieter surroundings for patients and students.
Smart City Administration	Uses data for urban planning and smart city development.
System Administrator	Maintains the software, cloud server, and IoT devices.

Importance of the Stakeholders

Each stakeholder contributes to the success of the system. Government agencies use the platform to enforce environmental regulations, industries ensure compliance with noise limits, citizens receive timely alerts, and administrators maintain the software infrastructure. Together, these stakeholders help improve urban environmental quality and public well-being.



2. Create Epics and User Stories

Epic 1: User Authentication

User Story 1

- **As a** System Administrator,
- **I want** to securely log into the system,
- **So that** I can manage users and monitor system activities.

User Story 2

- **As a** Municipal Officer,
- **I want** to log in to the dashboard,
- **So that** I can monitor noise pollution in my assigned area

Epic 2: Real-Time Noise Monitoring

User Story 3

- **As a** Pollution Control Officer,
- **I want** to view real-time noise levels,
- **So that** I can identify areas exceeding permissible limits.

User Story 4

- **As a** System Administrator,
- **I want** IoT sensors to continuously collect noise data,
- **So that** the system always has updated information.

Epic 3: GIS-Based Hotspot Detection

User Story 5

- **As a** Municipal Officer,
- **I want** to view noise hotspots on a GIS map,
- **So that** I can take immediate corrective actions.

User Story 6

- **As a** Traffic Police Officer,
- **I want** to identify high-noise traffic zones,
- **So that** traffic congestion can be reduced.

Epic 4: AI-Based Prediction

User Story 7

- **As a** Pollution Control Officer,
- **I want** the system to predict future noise trends,
- **So that** preventive measures can be planned in advance.

User Story 8

- **As a** Smart City Administrator,
- **I want** AI to analyze historical noise data,
- **So that** I can make better urban planning decisions.

Epic 5: Alerts and Notifications

User Story 9

- **As a** Municipal Officer,
- **I want** to receive instant alerts,
- **So that** immediate action can be taken when noise exceeds the limit.

User Story 10

- **As a** Citizen,
- **I want** to receive notifications about excessive noise,
- **So that** I can avoid highly polluted areas or report disturbances.

Epic 6: Reports and Analytics

User Story 11

- **As a** Pollution Control Board Officer,
- **I want** to generate monthly environmental reports,
- **So that** I can verify compliance with noise regulations.

User Story 12

- **As a** Smart City Administrator,
- **I want** to analyze historical noise trends,
- **So that** future environmental policies can be improved.

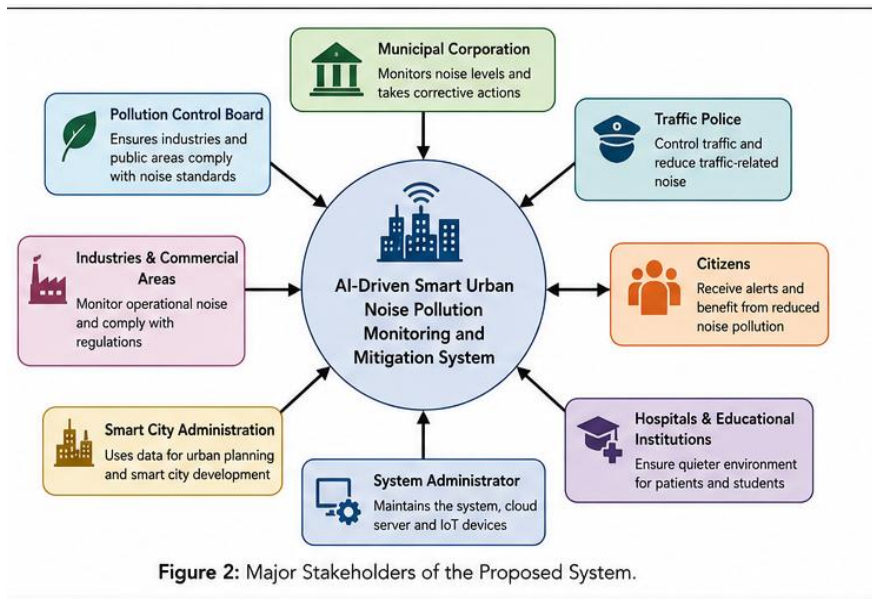
3. Define Acceptance Criteria

The following acceptance criteria ensure that each user story is completed successfully and meets the functional requirements of the system.

User Story	Acceptance Criteria (Given – When – Then)
User Login	Given the user has valid login credentials, When the user enters the username and password, Then the system should successfully log the user into the dashboard.
Real-Time Noise Monitoring	Given the IoT sensors are active, when noise data is collected, Then the system should display the current noise level in real time.
Noise Hotspot Detection	Given the system receives location-based noise data, When the noise exceeds the permissible limit, Then the GIS map should highlight the area as a hotspot.
AI Prediction	Given historical and current noise data are available, When the Machine Learning model analyzes the data, Then it should predict future noise trends accurately.
Alert Notification	Given the detected noise level exceeds the predefined threshold, When the system processes the data, Then an alert notification should be sent to the concerned authority immediately.
Report Generation	Given the monitoring data is stored in the database, When the user requests a report, Then the system should generate a detailed environmental compliance report.
Dashboard Visualization	Given valid sensor data is available, When the dashboard is opened, Then graphs, charts, and GIS maps should display the latest information.
Citizen Notification	Given excessive noise is detected in a locality, When the system generates an alert, Then nearby registered citizens should receive a notification

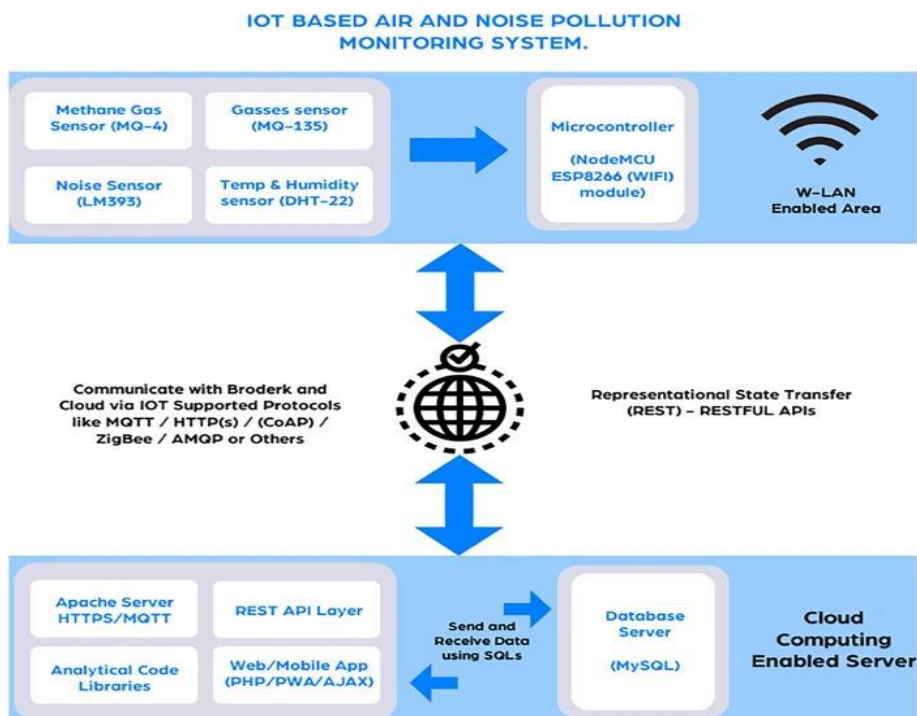
4. Product Backlog Prioritization

The Product Backlog is a prioritized list of features and functionalities required to develop the AI-Driven Smart Urban Noise Pollution Monitoring and Mitigation System. Features are prioritized based on business value, user needs, and project dependencies.



Priority	Product Backlog Item	Description	Sprint
High	User Login & Authentication	Secure login for authorized users	Sprint 1
High	IoT Sensor Integration	Collect real-time noise data	Sprint 1
High	Database Management	Store and manage sensor data	Sprint 1
High	Real-Time Noise Monitoring	Display live noise levels	Sprint 1
Medium	GIS Noise Hotspot Mapping	Display hotspots on digital maps	Sprint 2
Medium	AI/ML Noise Prediction	Predict future noise trends	Sprint 2
Medium	Alert & Notification System	Send alerts when noise exceeds limits	Sprint 2
Medium	Dashboard & Data Visualization	Display graphs and charts	Sprint 2

Low	Report Generation	Generate daily/monthly reports	Sprint 3
Low	Citizen Feedback Module	Allow citizens to report complaints	Sprint 3



5. Sprint Backlog (Sprint 1)

Sprint Goal

Develop the core functionalities required for collecting and displaying real-time noise pollution data.

Sprint Duration

2 Weeks

Task	Assigned To	Status
Design Login Page	Developer	To Do
Develop User Authentication	Developer	To Do
Connect IoT Sensors	IoT Engineer	To Do
Create Database Tables	Database Administrator	To Do
Develop Real-Time Monitoring Module	Developer	To Do
Unit Testing	Tester	To Do

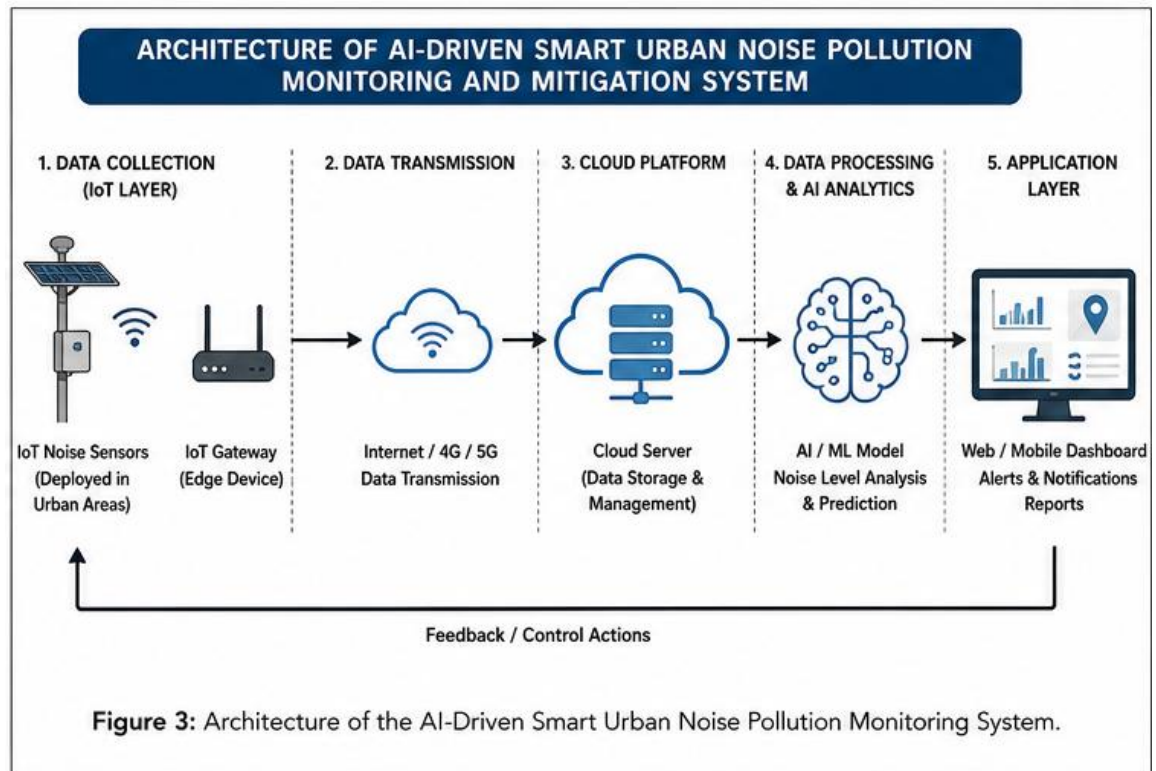
Sprint Deliverables

At the end of Sprint 1, the following features will be completed:

- User Login Module
- IoT Sensor Integration
- Database Setup
- Real-Time Noise Monitoring
- Initial Testing

6. Story Point Estimation

Story Point Estimation is used in Agile to estimate the effort, complexity, and time required to complete each user story. The **Fibonacci sequence (1, 2, 3, 5, 8, 13)** is commonly used because it provides a realistic estimation of work.



User Story	Story Points	Reason
User Login & Authentication	3	Simple user authentication
IoT Sensor Integration	13	Complex hardware and software integration
Database Management	5	Database design and storage
Real-Time Noise Monitoring	8	Continuous sensor data processing
GIS Hotspot Mapping	8	Map integration and visualization
AI/ML Noise Prediction	13	Machine learning model development
Alert & Notification System	5	Automatic alert generation
Dashboard & Data Visualization	8	Graphs, charts, and reports
Report Generation	3	Generate PDF/Excel reports
Citizen Feedback Module	5	Complaint submission and tracking

7. Sprint Goal & Expected Deliverables

Sprint Goal

The primary goal of the first sprint is to develop the core functionalities of the AI-Driven Smart Urban Noise Pollution Monitoring and Mitigation System. This includes implementing secure user authentication, integrating IoT sensors for real-time data collection, setting up the database, and displaying live noise monitoring data on a dashboard.

Expected Deliverables

At the end of Sprint 1, the team is expected to deliver:

- A secure user login and authentication system.
- IoT sensor integration for real-time noise data collection.
- A centralized database for storing sensor data.
- A real-time dashboard displaying current noise levels.
- Initial unit testing of all implemented modules.
- Documentation of Sprint 1 features and testing results.

Benefits of Agile in this Project

The Agile methodology offers several advantages for this project:

- Enables faster development through iterative sprints.



- Allows continuous feedback from stakeholders.
- Improves collaboration among developers, testers, and clients.
- Makes it easier to add new features in future sprints.
- Ensures early identification and correction of defects.
- Increases software quality through continuous testing and review.

Conclusion

The AI-Driven Smart Urban Noise Pollution Monitoring and Mitigation System applies Agile principles to develop a reliable, scalable, and intelligent solution for monitoring urban noise pollution. By using well-defined user stories, acceptance criteria, product backlogs, sprint planning, and story point estimation, the project ensures efficient software development while meeting stakeholder requirements. The system contributes to smarter urban management, improved environmental monitoring, and better public health.

